

Measure and Improve Your Food: Ingredient Estimation Based Nutrition Calculator

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Abstract

We developed an application that can easily calculate the nutritional content of a meal by utilizing our multimedia recipe dataset tied to the Nutrition Facts table and an ingredient estimation model. A CLIP-based image recognition model and an ingredient co-occurrence graph assist a user in selecting the appropriate ingredient from over 2,500 ingredients. Unlike traditional food applications, ours calculates nutrition from a list of ingredients, allowing the user to see how each ingredient affects the overall nutrition. The user adjusts the amount of each ingredient and knows how to change their meal to meet their body's needs.

CCS Concepts

• **Applied computing** → *Health care information systems*; • **Information systems** → *Multimedia content creation*.

Keywords

ingredient estimation, nutrition, dietary assessment application

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1 Introduction

Our motivation is to know if the nutrition of the meal I prepare is appropriate for me and, if not, how I can change the ingredients to make the meal fit my body's requirements. The nutritional content of a meal can be roughly calculated by adding up the nutritional content of its ingredients. However, this “ingredient” should not be free-form, as when writing a recipe, but should correspond to the “food item number” associated with the food nutrition facts. In many countries, such data on food nutrition facts is available

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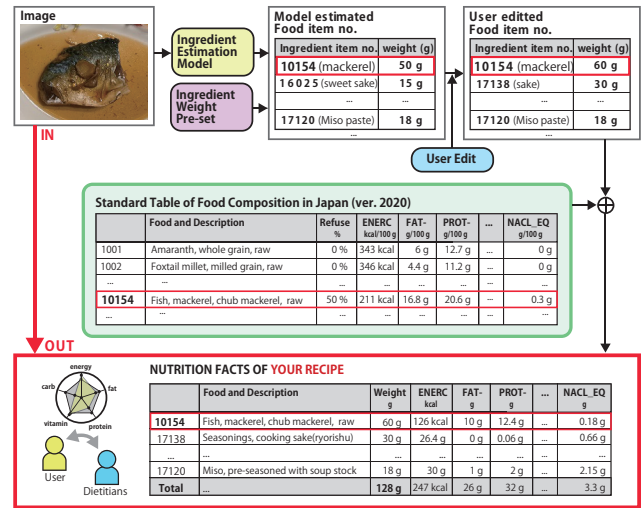


Figure 1: The flow of calculating nutritional value from a food image.

for common ingredients. In Japan, the Japanese government has published the Standard Tables of Food Composition in Japan [15]. Evaluating the nutritional content of a meal using this table offers an accurate and detailed dietary record in the same way that a dietitian evaluates a patient's dietary record to conduct a nutritional assessment. However, there are about 2,500 food choices, and it is difficult to pick the appropriate food item number for all ingredients. For example, “beef round” alone has 24 choices.

In our application, in addition to the image recognition model that estimates the ingredients as food item numbers from the meal image, a co-occurrence probability graph between ingredients enables rapid input of the ingredient list. Preset portions help a user to enter the weight of each ingredient. Importantly, a user can not only know the nutritional content of their current diet but can also adjust the portion of each ingredient to see how they can modify their current diet to adequately meet their body's needs. The ability to reflect an ingredient's weight in the nutrition graphs and tables in real time helps users improve their meals by adjusting the weight of an ingredient without changing the type of dish.

The ingredient estimation is performed on a cloud server and meal records are stored for each user in a database on the server. This data is used not only for users and their dietitians to look back on their food records, but also for future model improvements.

Table 1: Performance of ingredient estimation with different backbone

Backbone	Precision	Recall	R@5	R@10	Micro-F1
ResNet50[17]	53.0%	47.5%	41.6%	58.7%	50.1%
ResNet101[17]	52.5%	54.1%	43.9%	61.8%	53.3%
CLIP(Vit/32) (Ours)	55.4%	59.7%	46.2%	67.3%	55.8%

2 Related works

Several methods for estimating nutrition by recognizing meal images have already been studied on Recipe1M+, Nutrition5k, and nutrition-labeled proprietary datasets [4–6, 8, 11–14]. Their models estimate the total nutritional value of a meal, not by ingredient, and it is impossible to know which ingredient impacts each nutrient. High-precision ingredient estimation remains challenging, but the estimation results must be more accurate for practical applications, such as nutritional assessment by dietitians. It is reasonable to break down nutritional calculations to the ingredient level to ensure their accuracy and manually correct errors.

The nutrient values provided in the Recipe1M+ and Nutrition5k datasets are based on the USDA Food and Nutrient Database [16]. This provides data on the nutrient composition of foods consumed in the USA for macronutrients (e.g. carbohydrates, proteins and fats), micronutrients (including vitamins and minerals) and other food compounds. This is almost the same format as our application’s nutrient data, so it can be easily adapted, and we consider this our next task.

Nutrition calculation requires not only the food item numbers but also their weights. Depth or multi-view images have been reported to be useful in estimating an amount of a meal [1, 8, 9, 12, 14, 18]. However, for simplicity, we decided to use a single RGB image as input and preset the median weight per serving of each food item as the default weight.

3 System Process Flow (Figure. 1)

3.1 Backend

Ingredient Estimation:

In our previous work, we constructed a dataset of 1.6M recipes with images in the Cookpad+ dataset [2], linking their ingredient lists to 587 food item numbers and a label for “other” and weights in grams [17]. Here, the 587 food item numbers correspond to the top 2000 most frequently occurring ingredient expressions in the Cookpad+ dataset. While our previous method employed ResNet50 and ResNet101 [3] for ingredient estimation, recent studies have reported that Transformer-based backbones provide better performance [8, 12]. We also employed a CLIP-based model in this application to benefit from pre-training with large datasets [10]. We connected an MLP to the output by CLIP’s image encoder to link them to 588 food item numbers including “other” and trained on our dataset. The results of ingredient estimation are shown in Table 1, which demonstrates a 2.5% improvement in Micro-F1 compared to the CNN-based approach. Currently, the system is operated on a CPU cloud server, and recognition time is about 3 seconds.

Ingredient Co-occurrence Graph:

There is a bias in the combination of ingredients used simultaneously in one dish. For example, when “flour” is used, “butter”

and “eggs” are often used at the same time, and when the ingredient “pasta” is used, “bacon” is more likely to be used than other ingredients. Therefore, we calculated the co-occurrence probabilities among the 588 ingredients above based on all recipes in our dataset. We obtained a co-occurrence graph by setting the probability to 0 if the probability was lower than a threshold of 0.3. When a user selects an ingredient, the likelihoods of the ingredients not yet selected are weighted according to their co-occurrence probability with the selected ingredient, resulting in a change in the order of the candidate ingredients.

3.2 Frontend and User Experience

[Step 1] Set nutrition of body requirement:

A user sets sex, age, and physical activity level to obtain target values for energy, protein, fat, carbohydrate, and salt per day for that user based on the Japanese Nutritional Intake Standards [7]. Assuming the user eats three meals daily, the value divided by three is reflected in the nutrition graph as the target value per meal.

[Step 2] Upload meal image and input the ingredient list:

The user uploads a photo of a dish that shows only one item. The image recognition model described in Sec. 3.1 estimates the likelihood of each ingredient from that image and presents the top 10 items to the user.

Each time the user selects an ingredient, the co-occurrence graph described in Sec. 3.1 will reorder the list of possible ingredients that have not yet been selected. The “Generate new candidates” button replaces the unselected ingredient candidates with the next ten highest likelihood ones. If the ingredient the user wants to enter is not recommended or is not included in the 587 ingredients mentioned above, the user can enter it via text search.

[Step 3] Adjust weights from preset values:

After completing the ingredients input, a slider appears to adjust the weight for each ingredient. This slider can be adjusted from 0 to 2 times around the default value preset for each ingredient. Currently, a default value of each ingredient’s weight is its median weight in our recipe dataset, but it will also be obtained by image recognition in our subsequent work. The weight the user adjusts is immediately reflected in the graph and table.

[Step4] Explore ways to improve your meal more:

Once the user has reviewed the nutrition of the current meal, they can adjust the amount of each ingredient to bring the nutritional value closer to the target value. By seeing how changing the amount of any ingredient changes the total nutritional value, the user can learn how to change their meal to make it healthier. For example, they may find that reducing pasta and increasing meat can decrease carbohydrates without decreasing protein.

4 Conclusion

We developed an application that can rapidly calculate the nutritional content of a meal by utilizing our multimedia dataset tied to the Nutrition Facts table and an ingredient estimation model. Dietary needs are diverse. Everyone’s requirements are different, and the applications they require should be different. We make this application open source, and building a platform for dietary management applications is our future challenge.

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